

Tonglin YAN

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PhD student at Université Paris-Saclay working at the intersection of machine consciousness, multimodal machine learning, and virtual reality. My research focuses on integrating the computational consciousness model with large language models to study and simulate verbal and non-verbal social cognition in virtual agents. Strong background in AI, NLP, and immersive VR systems, with poster communications at major international consciousness conferences (ASSC).

Education

Université Paris-Saclay | Orsay, France Oct. 2023 – present
PhD in Computational Psychology

- Advisors: **David Rudrauf** (CIAMS, Université Paris-Saclay), **Alain Finkel** (LMF, ENS Paris-Saclay), **Grégoire Sergeant-Perthuis** (CQSB, Sorbonne Université)
- Dissertation: Integration of advanced multimodal machine learning models in the Projective Consciousness Model for the study of non-verbal and verbal social behaviors in virtual reality.
- Research topics: machine consciousness, multimodal LLMs, computational modeling of consciousness, VR-based user experience.

University of Rouen Normandy | Rouen, France Sept. 2021 – Sept. 2022
M.S. in Data Science and Engineering (Double degree with INSA Rouen Normandie)

INSA Rouen Normandie | Rouen, France Sept. 2017 – Sept. 2022
Engineering degree in Mathematical and Software Engineering

Professional and Research Experiences

Algorithm Engineer, GuanYun Technology Jan. 2023 – Jun 2023

- Designed and implemented an human motion capture system based on 9-axis IMU sensors.
- Developed signal processing and kinematic reconstruction algorithms for full-body motion estimation.
- Integrated motion data into Unity3D for real-time visualization and analysis in VR environments.

Research Assistant, ISIR, Sorbonne University Mar. 2022 – Sept. 2022

- Designed an end-to-end pipeline for constructing a biology dataset.
- Fine-tuned BERT and T5 models for data-to-text generation tasks, with a 15% improvement in accuracy compared to baseline models.
- Evaluated model scalability and performance across datasets.

Research Assistant, New York University June 2021 – Sept. 2021

- Predicted animal trajectories (ants/birds) using time-series analysis.
- Cleaned and imputed data via linear regression; extracted features with random forests.

Part-Time Assistant, Deloitte China July 2020 – Sept. 2020

- Conducted a case study on retail markdown strategies, analyzing promotional impacts.
- Delivered client recommendations via data-driven presentations.

Conferences and Workshops

- **Yan, T.**, Sergeant-Perthuis, G., Williford, K., Rudrauf, D. PCM-LLM: Bridging Non-Verbal Consciousness Modeling and Language Processing to Make Intelligent Social Virtual Agents Closer to Human Beings. **28th Annual Meeting of Association for the Scientific Study of Consciousness**, July 2025, Crete, Greece ([link to poster](#))
- Ruet, N., **Yan, T.**, Ognibene, D., Williford, K., Rudrauf, D., Sergeant-Perthuis, G. Exploring how the geometry of the representation space influences curiosity-based exploration. **27th Annual Meeting of Association for the Scientific Study of Consciousness**, July 2024, Tokyo, Japan. ([link to poster](#))

- Rudrauf, D., **Yan, T.**, Ruet, N., Williford, K., Sergeant-Perthuis, G. Integrated Information Theory (IIT) with Simple Maths. **27th Annual Meeting of Association for the Scientific Study of Consciousness**, July 2024, Tokyo, Japan. ([link to presentation](#))
- Sergeant-Perthuis, G., Ruet, N., **Yan, T.**, Williford, K., Rudrauf, D. Dualities in G-Spaces May Underly Pre-Reflective Self-Consciousness. **27th Annual Meeting of Association for the Scientific Study of Consciousness**, July 2024, Tokyo, Japan. ([link to poster](#))
- Zhang, L. X., **Yan, T.**, Takeda, K., Paganin, M. Optimal Transport of Hallucinatory Colors. **Post-ASSC Satellite Symposium**, July 2024, Kyoto, Japan. ([link to poster](#))

Publications

Under Review

1. **Yan, T.**, Sergeant-Perthuis, G., Rudrauf, D. A Hybrid PCM-LLM Architecture for Interpretable and Socially Situated Agents, 2026.
2. **Yan, T.**, Tsukahara, J., Williford, K., Rudrauf, D., Sergeant-Perthuis, G. Formalizing Consciousness: A Comparative Mathematical Analysis of Leading Theories, 2026

Preprints

4. **Yan, T.**, Sergeant-Perthuis, G., Williford, K., Rudrauf, D. Integrating Machine Consciousness Simulation and LLMs Toward Verbal and Non-Verbal General Intelligence in Artificial Agents, 2025.
5. **Yan, T.**, Sergeant-Perthuis, G., Rudrauf, D. PCM-LLMs: a hybrid architecture to enhance human-like social intelligence in virtual agents, 2025.
6. Sergeant-Perthuis, G., **Yan, T.**, Ruet, N., Williford, K., Rudrauf, D. Integrated Information Theory (IIT) with Simple Maths, 2024.

Academic Training

Qualia Structure Summer School | Osaka and Awaji, Japan

June 2024

- **Theme:** Explored interdisciplinary approaches to the structure of consciousness, integrating philosophy, cognitive science, neuroscience, mathematics, and artificial intelligence.

Supervision Experience

Co-supervised Yannick Zelle (M1 intern, ENS), with David Rudrauf

2025

Co-supervised Germain Poloudenny (M2 intern, Université d'Artois), with David Rudrauf

2024

Skills

Programming: Python (PyTorch, TensorFlow), C#, Java, C++, R

Technologies: Unity 3D

Languages: Mandarin (native), English (B2), French (B2)

Interests: Tennis, Calligraphy, Bouldering, Travel